Fruit Image Classifier with CNNs

1. Introduction

This project focuses on fruit classification using deep learning models. The chosen models include a Custom CNN, VGG16, and ResNet50. The dataset was reduced to 10 fruit classes for training and evaluation.

2. Problem Statement

Manual classification of fruits is time-consuming and prone to errors. The project aims to automate fruit classification using image recognition models, improving accuracy and efficiency.

3. Project Objectives

- Build and train a Custom CNN, VGG16, and ResNet50 on fruit dataset
- Compare their performance
- Test with unseen images
- Provide visualization of results

4. Methodology

Dataset: 4869 images across 10 fruit classes (Banana, Cherry, Cocos, Kiwi, Mango, Peach, Lemon, Pineapple, Watermelon, Strawberry)
Training: 3896 images
Validation: 973 images
Testing: 1633 images
Tools: TensorFlow, Keras, Matplotlib
Models: Custom CNN, VGG16 (Transfer Learning), ResNet50 (Transfer Learning)

5. Implementation

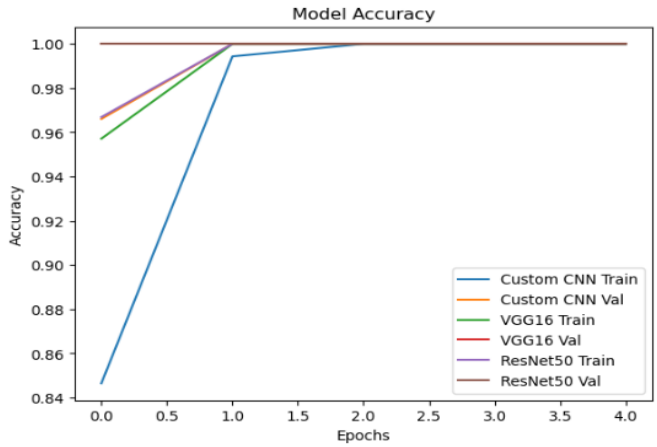
Step 1: Dataset preprocessing (resizing to 128x128, normalization, augmentation)
Step 2: Model creation
Step 3: Training with categorical cross entropy loss & Adam optimizer
Step 4: Validation & Testing
Step 5: Visualization of accuracy/loss curves, confusion matrix, bar plots

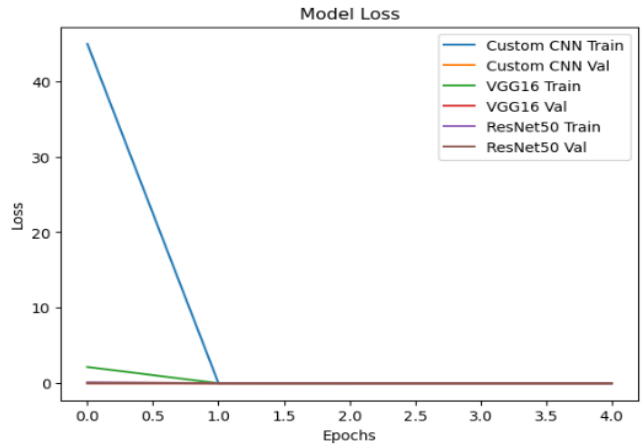
6. Demo / Working

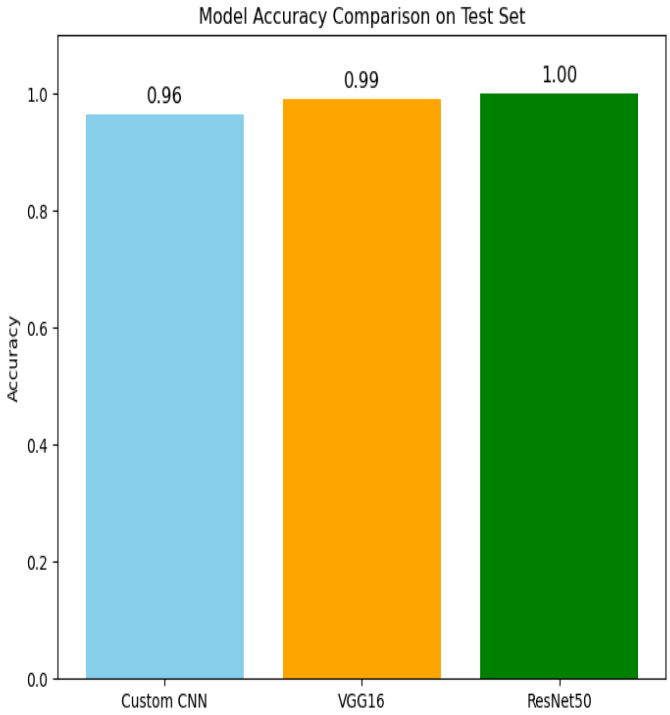
Prediction Example:
Custom CNN → Predicted: Banana 1 (confidence: 0.12)
VGG16 → Predicted: Banana 1 (confidence: 0.53)
ResNet50 → Predicted: Banana 1 (confidence: 0.23)

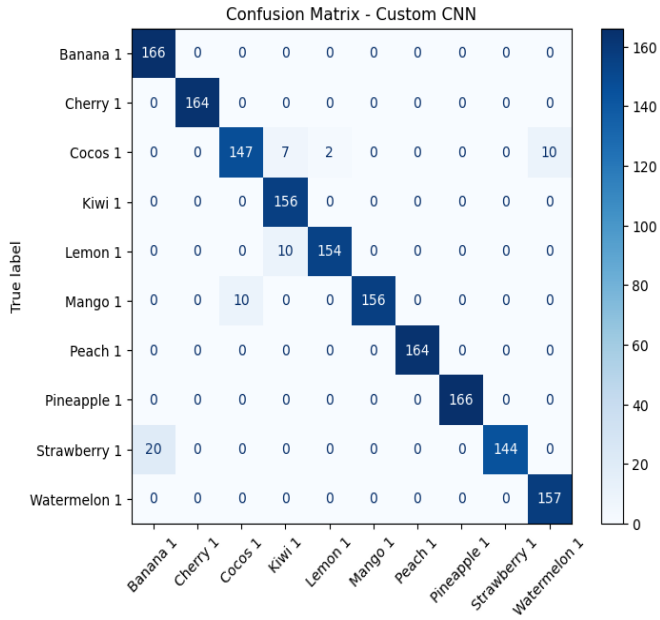
7. Results & Observations

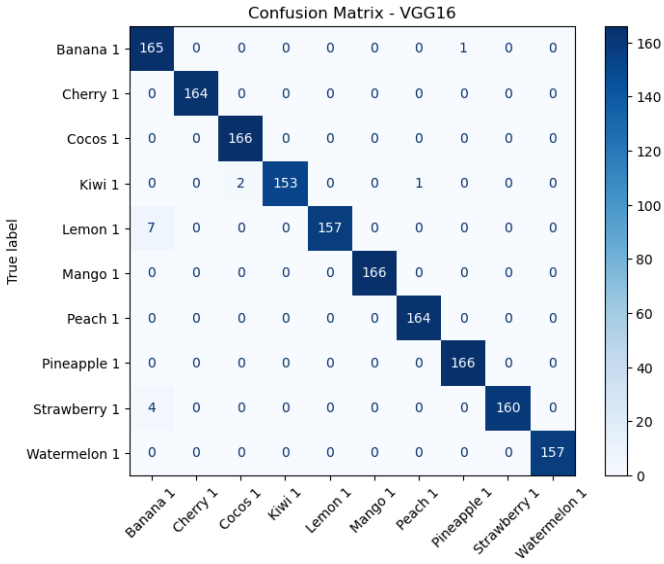
Final Model Comparison:
Custom CNN – 0.9638
VGG16 – 0.9908
ResNet50 – 1.0000

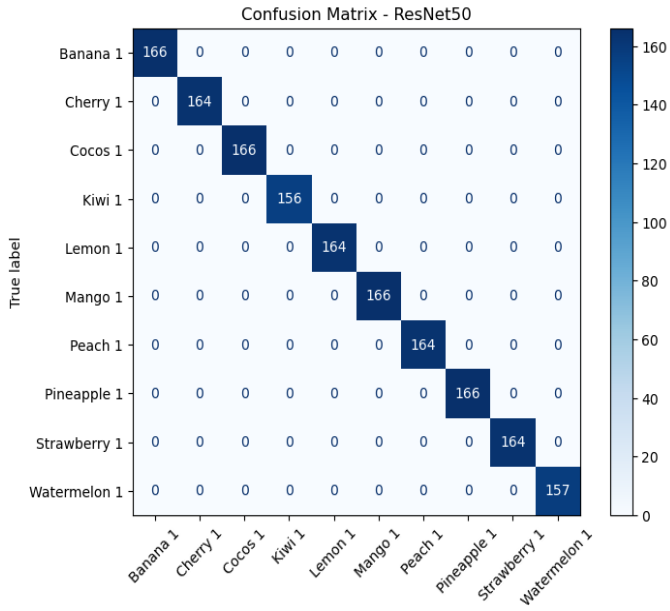












8. Conclusion & Future Work

The project achieved high accuracy with all three models. ResNet50 performed best with 100% accuracy. Future work may include:

- Expanding dataset with more fruit classes
- Deploying model in mobile or web apps
- Real-time fruit recognition system